

Real-Time Facial Emotion Recognition for Meeting Insights

A CNN-powered **Meeting Insight Recorder** Chrome extension

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Problem Statement

Online meetings are everywhere — but speakers fly blind.



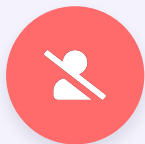
Meetings went digital

Presentations and interviews now happen mostly online.



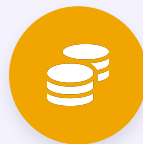
No feedback loop

Speakers cannot see how confident or clear they appear.



Self-judgement is hard

It is very hard to rate your own tone and emotions.



Coaching is costly

Human communication coaches are expensive and rare.

02 Objectives



Record meetings

01

A Chrome extension that captures meeting audio and video.



Detect emotions

02

Use a CNN to recognise facial emotions in real time.



Analyse speech

03

Transcribe the audio and measure communication quality.



Coach the user

04

Generate friendly AI feedback with clear, useful scores.

03

Literature Review

FER2013 was introduced in the ICML 2013 challenge (Goodfellow et al.) — 35,887 face images across 7 emotions.

~65%

Human accuracy

75.2%

Kaggle winner

73.3%

Tuned VGGNet



Research Gap

Most existing models stop at classifying emotions. They do not turn those emotions into practical, easy-to-read feedback that helps a person improve how they present in a meeting.

04

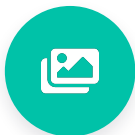
Dataset Description

Source: Kaggle — FER2013 (msambare)



35,887

total images



28,709 / 7,178

train / test



7 classes

emotion labels



48×48

grayscale pixels

The 7 emotion classes

Angry

Disgust

Fear

Happy

Neutral

Sad

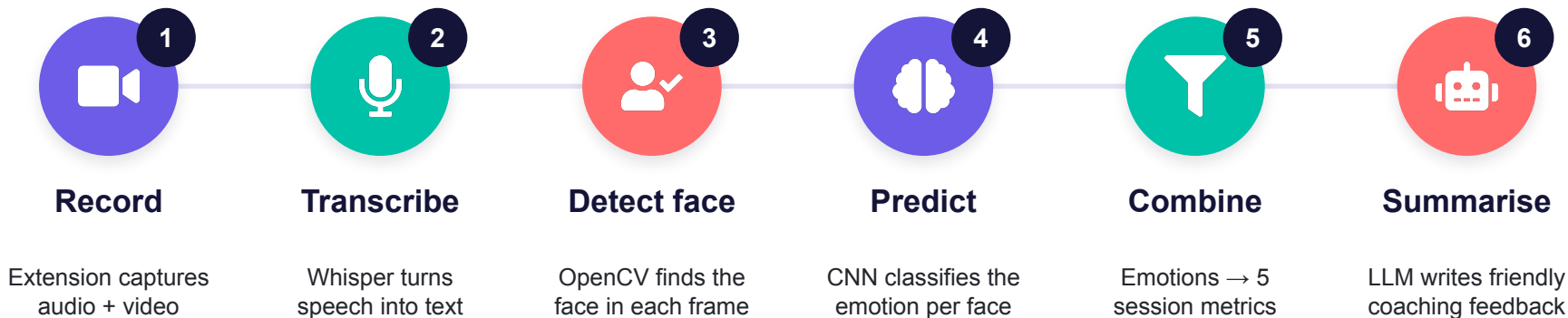
Surprise

Preprocessing: rescale pixel values to 0–1 → convert to grayscale → resize every image to 48×48.

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Methodology & System Workflow

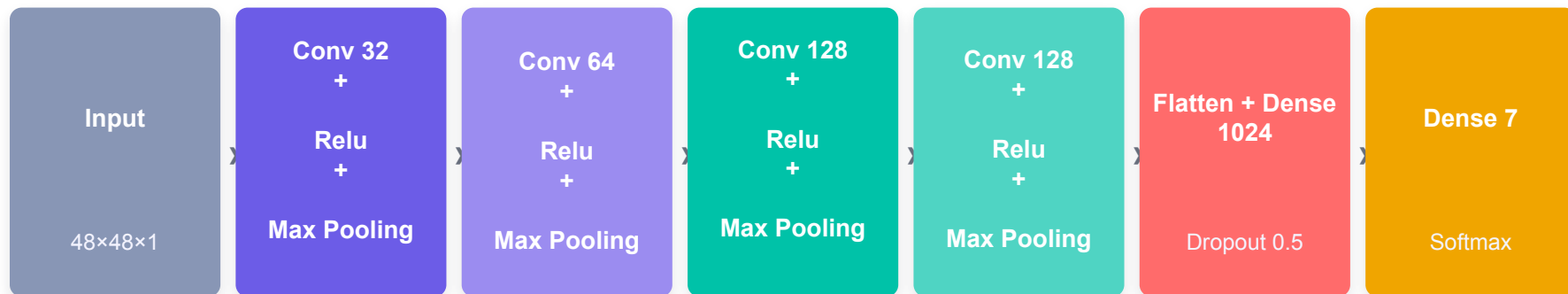
How a recorded meeting becomes coaching feedback — step by step.



AI techniques: CNN (image classification) • OpenCV Haar-cascade face detection • Whisper ASR • LLM summarisation

06

Model Architecture — CNN

**~2.3M**

parameters

**Adam**

+ exp. LR decay

**7**

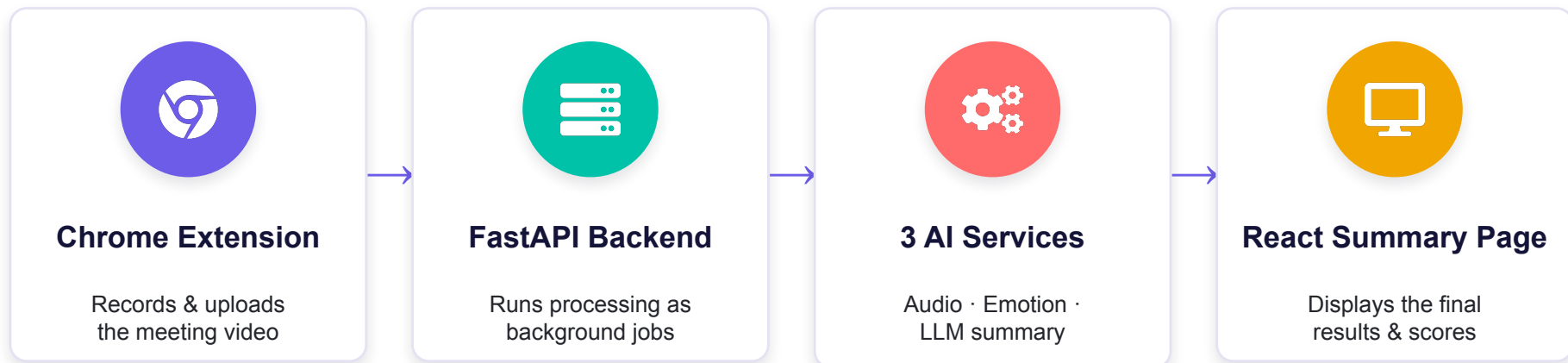
output classes

**categorical**

cross-entropy loss

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System Design — End-to-End Pipeline



Emotions are mapped into 5 session metrics

Confidence

Nervousness

Engagement

Frustration

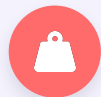
Emotional Stability

08 Experimental Results



65%

final test accuracy



1.13

final test loss



30

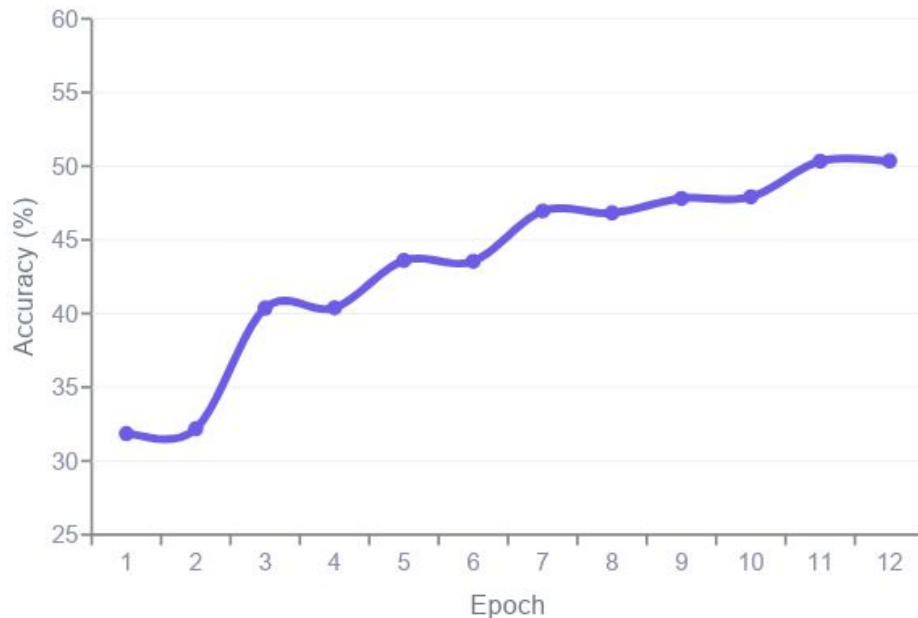
training epochs



64

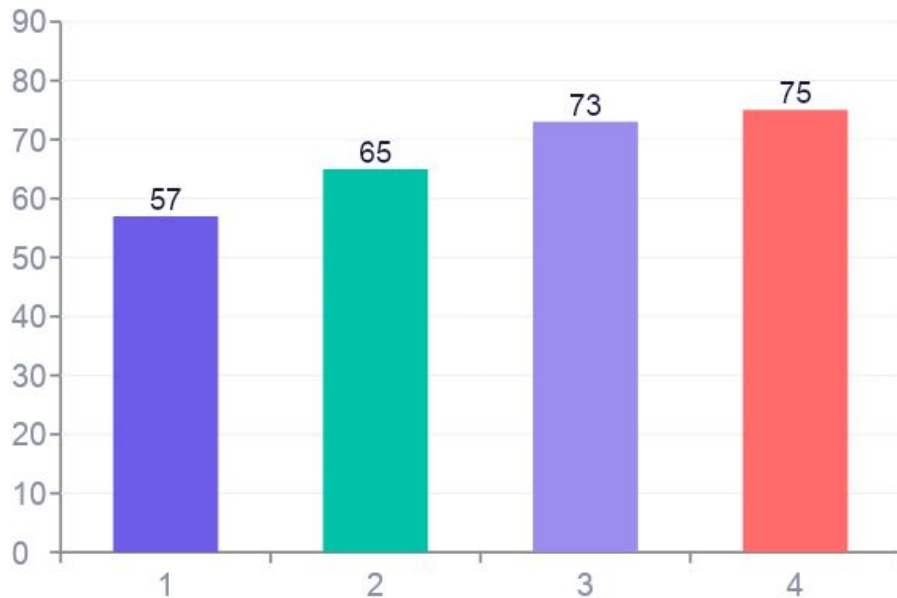
batch size

Validation accuracy climbing during training



Note: add your confusion matrix image here from the notebook.

Our model vs. published FER2013 results



Fair for a simple model

57% sits just below the ~65% human baseline — solid for a light 4-conv CNN vs deep VGG/ResNet models.



FER2013 is hard

Noisy labels and imbalanced classes limit accuracy for everyone using this dataset.



Our real value

We turn raw emotions into practical meeting coaching — not just a class label.

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Limitations & Future Work



Current Limitations

- Simple architecture, no data augmentation caps accuracy
- Class imbalance — very few 'disgust' samples
- A data-generator issue cut the effective training per epoch
- Emotion → metric weights are hand-set, not yet validated



Future Work

- Add data augmentation (flip, rotate, brightness)
- Use transfer learning with VGG or ResNet
- Balance classes / apply class weights
- Fix the generator so all 30 epochs train fully
- Validate the emotion-to-metric mapping on real data

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Individual Contributions



Niluminda

CNN Model

Trained the emotion CNN on the FER2013 dataset.



Buddhika

Backend / API

Built the FastAPI backend and REST API integration.



Oshini

Extension UI

Built the Chrome extension popup and recorder interface.



Irasha

Summary Page

Built the summary results page and feedback display.

Thank You

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